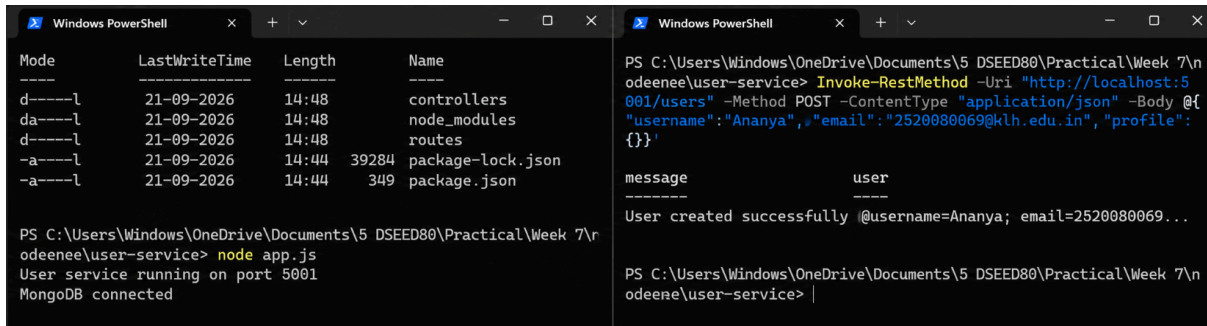


WEEK-7 PRACTICAL: Node.js User Service

Step – 1 Create a User — POST



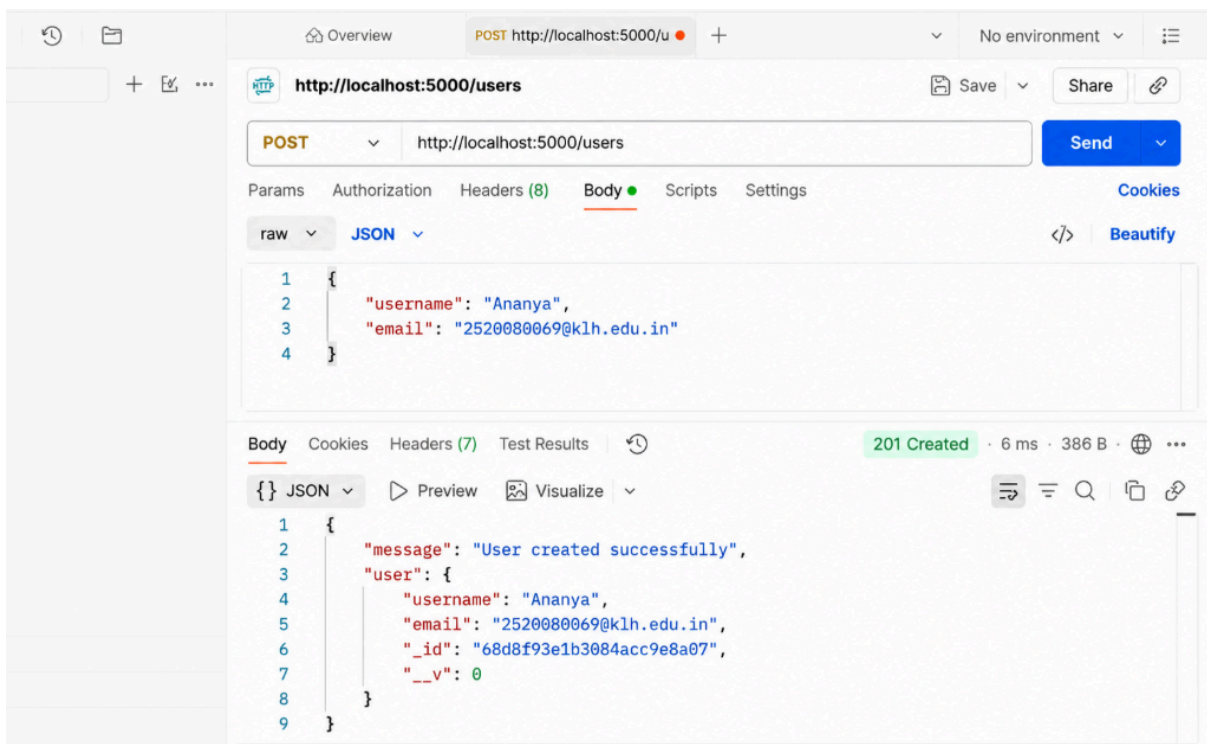
```
PS C:\Users\Windows\OneDrive\Documents\5 DSEED80\Practical\Week 7\odeenee\user-service> node app.js
User service running on port 5001
MongoDB connected

Mode      LastWriteTime         Length      Name
-----
d-----l  21-09-2026         14:48      controllers
da-----l  21-09-2026         14:48      node_modules
d-----l  21-09-2026         14:48      routes
-a-----l  21-09-2026        14:44    39284 package-lock.json
-a-----l  21-09-2026         14:44      349 package.json

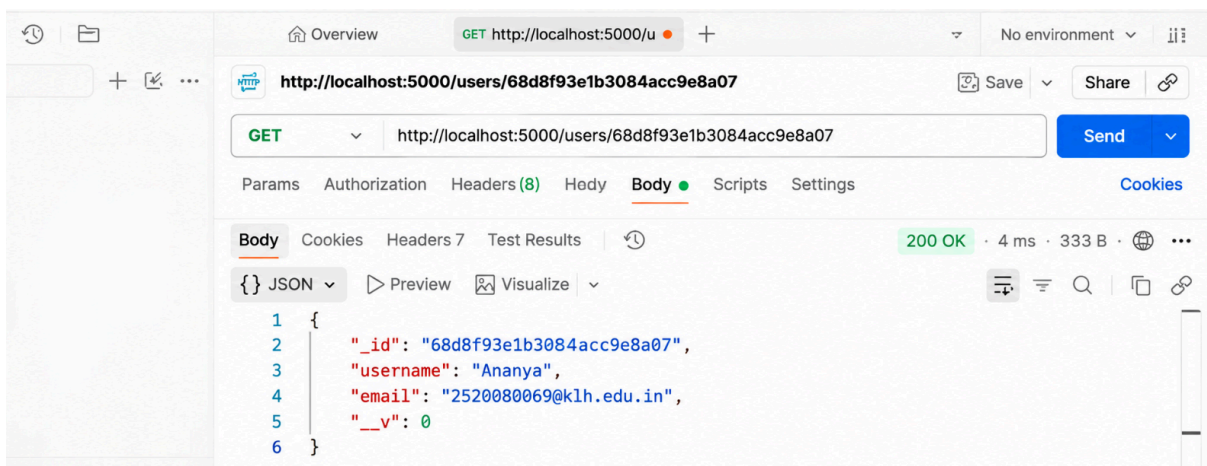
PS C:\Users\Windows\OneDrive\Documents\5 DSEED80\Practical\Week 7\odeenee\user-service> Invoke-WebRequest -Uri "http://localhost:5001/users" -Method POST -ContentType "application/json" -Body @{"username": "Ananya", "email": "2520080069@klh.edu.in", "profile": {}}
message      user
-----
User created successfully @username=Ananya; email=2520080069...

PS C:\Users\Windows\OneDrive\Documents\5 DSEED80\Practical\Week 7\odeenee\user-service> |
```

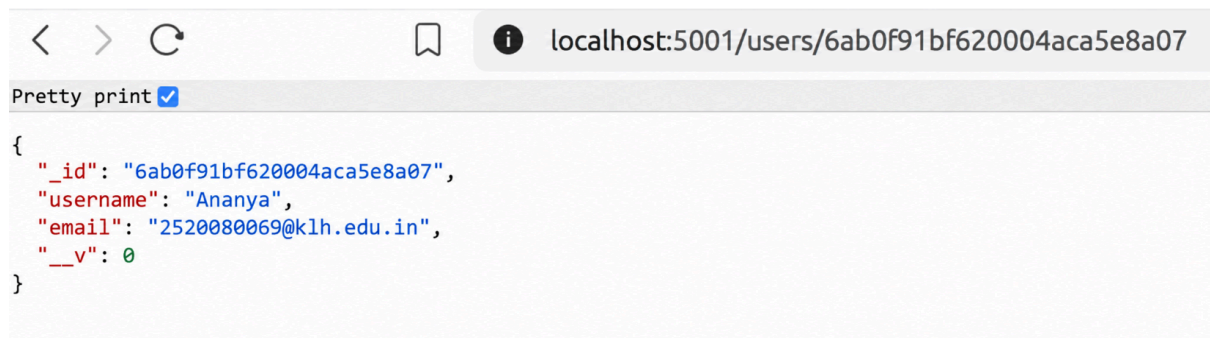
POST in mongosh:



Step 2: Get User — GET



Output on browser by id:

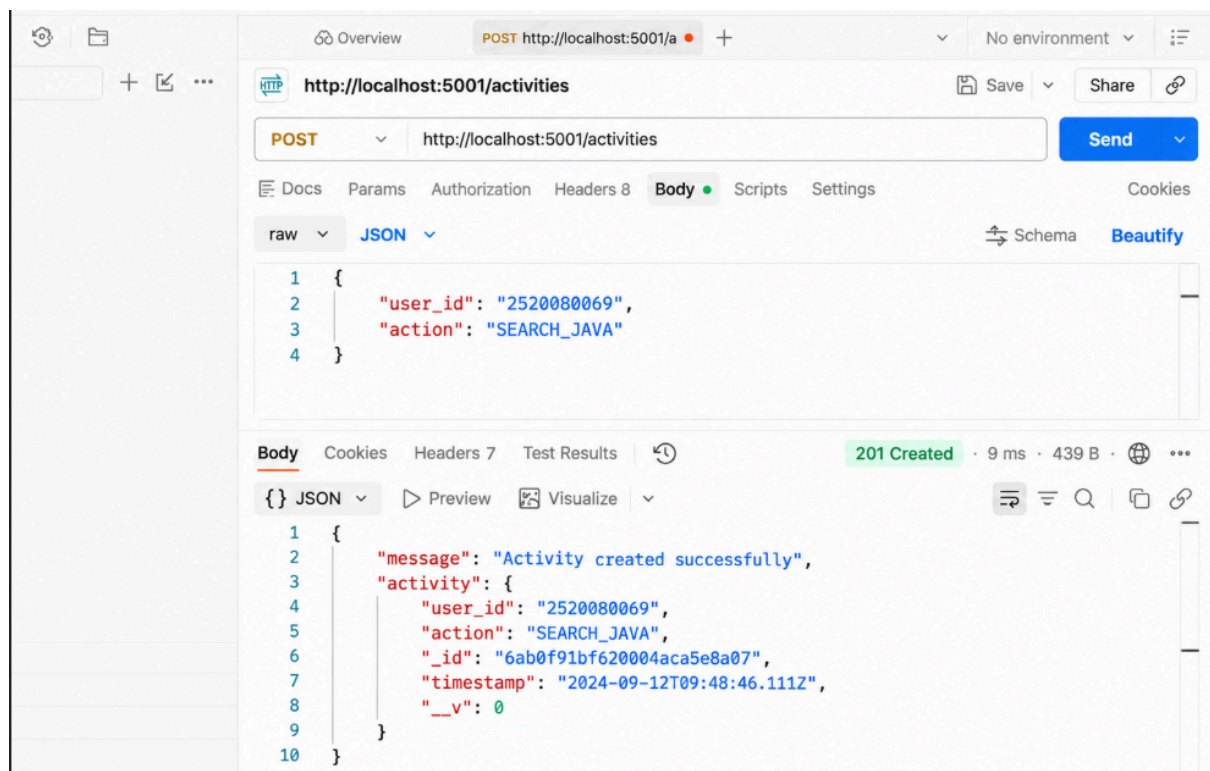


Step 3: Create Activity — POST

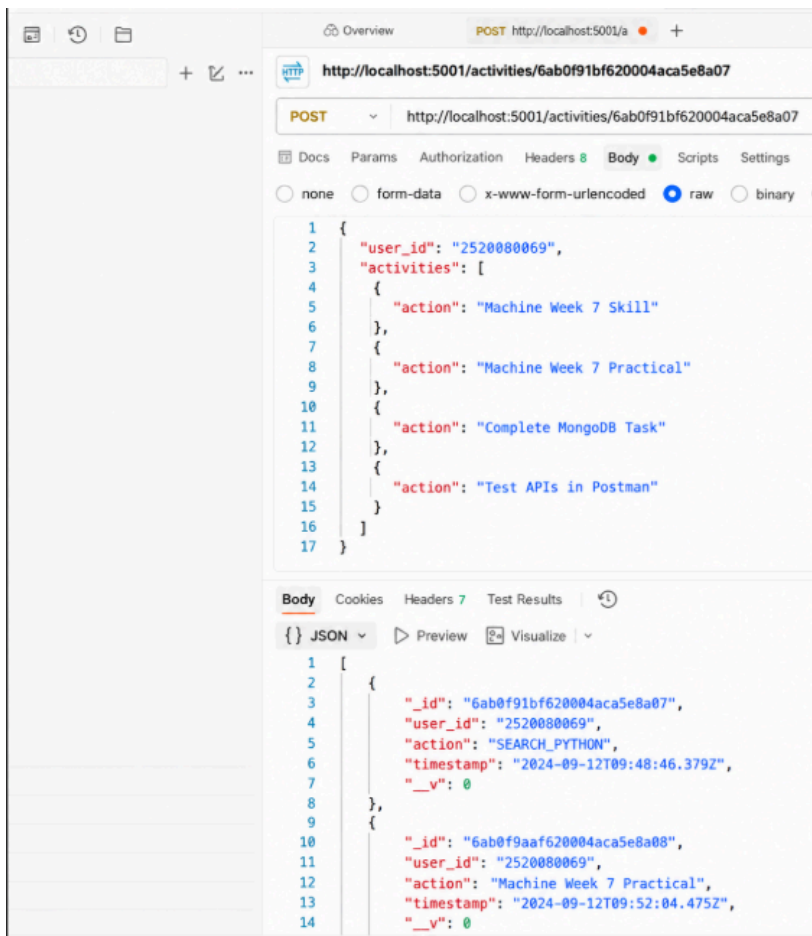
```
PS C:\Users\DELL\OneDrive\Documents\S6 EEE5208\Practical\Week 7\nodeeeee\user-services> Invoke-WebRequest -Uri "http://localhost:5001/activities" -Method POST -ContentType "application/json" -Body '{"user_id": "6ab0f91bf620004aca5e8a07", "action": "SEARCH_PYTHON"}'

message          activity
-----
Activity created successfully {user_id=6ab0f91bf620004aca5e8a07, action=SEARCH_PYTHON}
```

In mongosh:



Step 4: Add More Activities



```
PS C:\Users\DELL\OneDrive\Document\S6 EE5208\Practical\Week 7\nodeeeee\user-services> Invoke-RestMethod -Uri "http://localhost:5001/activities/6ab0f91bf620004aca5e8a07" -Method GET
```

```
_id      : 6ab0f915f620004aca5e8a8c
user_id  : 2520080069
action   : SEARCH_PYTHON
timestamp : 2024-09-12T09:48:46.379Z
__v      : 0

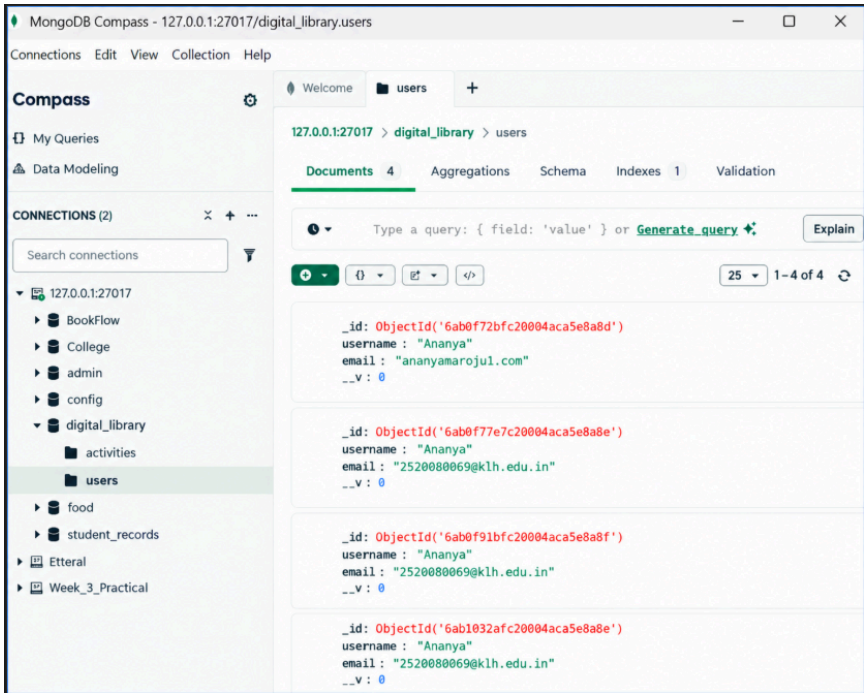
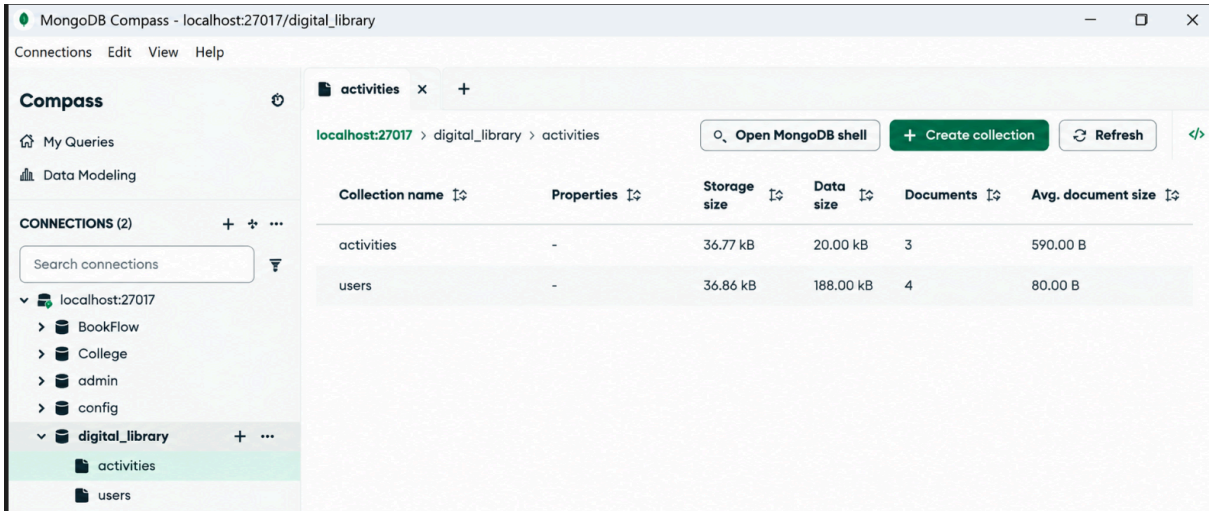
_id      : 6ab0f9aaf620004aca5e8a89
user_id  : 2520080069
action   : Machine Week 7 Practical
timestamp : 2024-09-12T09:52:04.475Z
__v      : 0

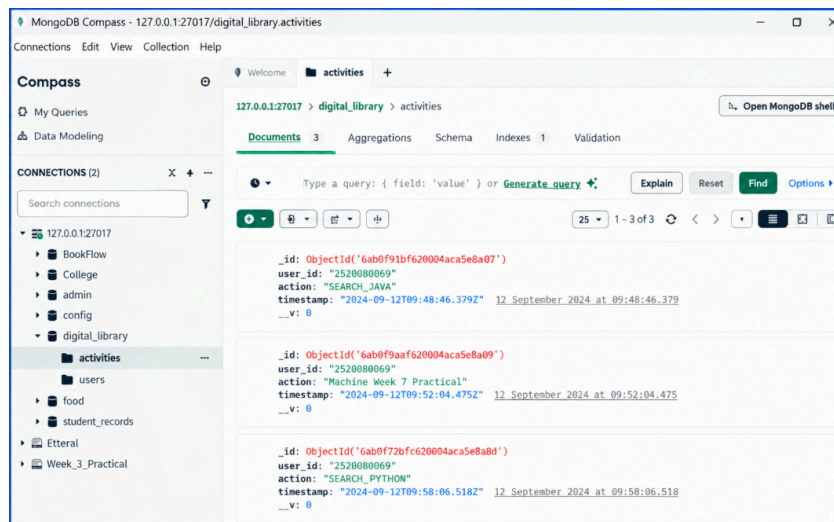
_id      : 6ab0f9bcf620004aca5e8a8e
user_id  : 2520080069
action   : SEARCH_JAVA
timestamp : 2024-09-12T09:58:06.518Z
__v      : 0
```

Output on browser:

```
localhost:5001/activities/6ab0f91bf620004aca5e8a07
Pretty print
[
  {
    "_id": "6ab0f915f620004aca5e8a8c",
    "user_id": "2520080069",
    "action": "SEARCH_PYTHON",
    "timestamp": "2024-09-12T09:48:46.379Z",
    "__v": 0
  },
  {
    "_id": "6ab0f9aaf620004aca5e8a89",
    "user_id": "2520080069",
    "action": "Machine Week 7 Practical",
    "timestamp": "2024-09-12T09:52:04.475Z",
    "__v": 0
  },
  {
    "_id": "6ab0f9bcf620004aca5e8a8e",
    "user_id": "2520080069",
    "action": "SEARCH_JAVA",
    "timestamp": "2024-09-12T09:58:06.518Z",
    "__v": 0
  }
]
```

MongoDB connection:





Folder Structure:

Week 7

|__nodememo

|__user-service

|

|__controllers

| |__activityController.js

| |__userController.js

|

|__routes

| |__activityRoutes.js

| |__userRoutes.js

|

|__app.js

|__db.js

|__package.json

|___.env

db.js

```
const mongoose = require("mongoose");
```

```
const MONGO_URI =
  process.env.MONGO_URI ||
```

```

    "mongodb://127.0.0.1:27017/digital_library";

mongoose.connect(MONGO_URI)
  .then(() => console.log("MongoDB connected"))
  .catch(err => console.error("MongoDB connection error:", err));

const userSchema = new mongoose.Schema({
  username: {
    type: String,
    required: true
  },
  email: {
    type: String,
    required: true
  },
  profile: {
    type: mongoose.Schema.Types.Mixed,
    default: {}
  }
});

const activitySchema = new mongoose.Schema({
  user_id: {
    type: mongoose.Schema.Types.ObjectId,
    required: true,
    ref: "User"
  },
  action: {
    type: String,
    required: true
  },
  timestamp: {
    type: Date,
    default: Date.now
  }
});

const User = mongoose.model("User", userSchema);
const Activity = mongoose.model("Activity", activitySchema);

module.exports = { User, Activity };

```

controllers/userController.js

```
const { User } = require("../db");

const createUser = async (req, res) => {
  try {
    const { username, email, profile } = req.body;

    const user = await User.create({
      username,
      email,
      profile
    });

    res.status(201).json({
      message: "User created successfully",
      user
    });
  } catch (error) {
    res.status(500).json({
      message: "Error creating user",
      error: error.message
    });
  }
};

const getUser = async (req, res) => {
  try {
    const user = await User.findById(req.params.id);

    if (!user)
      return res.status(404).json({
        message: "User not found"
      });

    res.json(user);
  } catch (error) {
    res.status(500).json({
      message: "Error getting user",
      error: error.message
    });
  }
};
```

```
module.exports = { createUser, getUser };  
controllers/activityController.js
```

```
const mongoose = require("mongoose");  
const { Activity } = require("../db");
```

```
const createActivity = async (req, res) => { try {  
  const { user_id, action } = req.body;  
  
  if (!mongoose.Types.ObjectId.isValid(user_id)) return  
    res.status(400).json({  
      message: "Invalid user_id"  
    });  
  
  const activity = await Activity.create({  
    user_id,  
    action  
  });  
  
  res.status(201).json({  
    message: "Activity created successfully",  
    activity  
  });  
} catch (error) {  
  res.status(500).json({  
    message: "Error creating activity",  
    error: error.message  
  });  
}  
};
```

```
const getActivities = async (req, res) => { try {  
  const { user_id } = req.params;  
  
  if (!mongoose.Types.ObjectId.isValid(user_id)) return  
    res.status(400).json({  
      message: "Invalid user_id"  
    });  
  
  const activities = await Activity  
    .find({ user_id })  
    .sort({ timestamp: -1 });
```

```

        res.json(activities);
    } catch (error) {
        res.status(500).json({
            message: "Error getting activities",
            error: error.message
        });
    }
};

module.exports = { createActivity, getActivities };

```

routes/userRoutes.js

```

const express = require("express");

const router = express.Router();

const {
    createUser,
    getUser
} = require("../controllers/userController");

router.post("/users", createUser);
router.get("/users/:id", getUser);

module.exports = router;

```

routes/activityRoutes.js

```

const express = require("express");

const router = express.Router();

const {
    createActivity,
    getActivities
} = require("../controllers/activityController");

router.post("/activities", createActivity);
router.get("/activities/:user_id", getActivities);

module.exports = router;

```

app.js

```
const express = require("express");

const app = express();

require("./db");

app.use(express.json());
const activityRoutes = require("./routes/activityRoutes"); const
userRoutes = require("./routes/userRoutes");

app.use("/", activityRoutes);
app.use("/", userRoutes);

app.listen(5001, () => {
  console.log("User service running on port 5001"); });
```